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A review of qualitative case methods trends and themes used in technology transfer research

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Abstract The focus of this paper is to review the qualitative case methods that have been used in technology transfer research over the last 20 years from 1996 to 2015. Case methods allow for more in-depth analyses and provide the opportunity to place research into a certain context due to the selection of e.g. specific sectors, institutions, countries, etc. Using a systematic literature review of five of the top journals in the field of technology transfer research, namely *Journal of Technology Transfer*, *Research Policy*, *Science and Public Policy*, *R&D Management and Technovation*, it yielded 107 articles using the search terms: "Technology Transfer" AND ("Case Study" OR "Case Method" OR "Qualitative"). Our findings indicate a clustering of themes using qualitative case methods around technology transfer mechanisms and TTOs, academic entrepreneurship, university-industry collaboration, commercialization as well as R&D and firm knowledge transfer. We also identify trends in case method technology transfer research with respect to authorship, location of papers, sectoral contexts, data collection, numbers of cases and data analysis software. We conclude our paper discussing the implications of trends and themes and suggest that researchers need to reflect on used terminology and their utilization and postulate a need for more plurality of data collection methods.

Keywords Technology transfer · Case method · Case study · Qualitative · Research methods · Data collection · Cases · Commercialization · Academic entrepreneurship

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JEL Classification C42 · O31 · O32 · O33 · O34 · L6 · L7 · L8 · L9

1 Introduction

Technology transfer is a rapidly growing field and this has resulted in a growing empirical base that has shaped policy and practice (see Bozeman 2000; Debackere and Veugelers 2005; Etzkowitz 2016; Grimaldi et al. 2011; Link et al. 2011; Shane 2004; Siegel et al. 2007). The field has focused on a variety of themes and actors in different institutional and country settings (see Bozeman et al. 2013). At a macro level, a growing body of research has focused on issues such as the triple helix (Leydesdorff and Etzkowitz 1996), entrepreneurial ecosystems (Wright 2014), national systems of innovation (Acs et al. 2016) and new emerging themes, such as public sector entrepreneurship (Leyden and Link 2015). More intense theoretical and empirical focus has been placed on understanding the various facets surrounding successful technology transfer between universities and industry (see Perkmann et al. 2013). This research has unearthed some of the antecedent characteristics and the benefits of technology transfer and has reinforced the interest on the different technology transfer mechanisms (Siegel et al. 2004) and corresponding technology transfer effectiveness (Bozeman et al. 2015). A growing body of literature is developing on the role of individual actors such as scientists, venture capitalists, consultants etc. in the technology transfer process and how these actors can shape and influence the technology transfer process (Audretsch et al. 2016; Cunningham et al. 2016). Some recent micro level research on technology transfer has focused on barriers and stimulants and motivations of different actors as well as on the influence of individual actors on the technology transfer process (Cunningham et al. 2014, 2015; Mangematin et al. 2014; Menter 2016; O'Kane et al. 2015b).

In the development of the research field related to technology transfer, a variety of methodological approaches have been used; however, there has been no systematic examination or reflective analysis on the research methods used that have advanced research in this field. To date, there has been no reflective focus on the methodological approaches—quantitative or qualitative—that researchers have used to study technology transfer. Such methodological reflects are common in other fields such as international entrepreneurship (see Coviello and Jones 2004) and management (see Scandura and Williams 2000). Given the growth of the technology transfer research over the last three decades, there is now a need to understand the underpinning methodological approaches that have been used to advance the technology transfer field, particularly with respect to qualitative case methods. Especially in growing fields of research such as technology transfer, considerations about the adequate usage of methodological approaches and terminology as well as the general scope of research are crucial to progress within the field and that his is underpinned with robust methodological approaches. Moreover, an overview of recent trends and themes thereby enables researchers to work on relevant research fields offering further insights into previously neglected areas of research, yet referring to previous work within the field. Such a review also provides researchers with an opportunity to reflect on and shape their own methodological approaches within the field and in doing so consider such questions as data collection approaches and sample sizes.

Using a systematic literature review of five of the top journals in the field of technology transfer, namely (listed in alphabetical order) *Journal of Technology Transfer, R&D Management, Research Policy, Science and Public Policy, R&D* and *Technovation*, we

identified studies utilizing qualitative case methods over the last 20 years from 1996 through 2015. The purpose of our paper is to examine qualitative case methods trends and themes used in technology transfer research. It also provides an opportunity to further critically reflect on the application of qualitative case methods in this field. Furthermore, we aim at stimulating the reflection of case methods terminology, their utilization as well as of data collection methods. Finally, we want to emphasize the general increase of qualitative case methods in technology transfer research and guide and encourage further research in this direction.

Our paper is organised as follows. In Sect. 2, our literature review focuses on case methods in terms of definitions, benefits and previous studies in other fields of research. In Sect. 3, we outline the methods that we used in our study and in Sect. 4, we present our findings. We conclude the paper with a discussion of our findings and outline the main implications for the field of technology transfer research with respect to the consistent usage of terminology, the plurality of data collection methods as well as qualitative and quantitative research designs.

2 Literature review

2.1 Qualitative case methods research studies in other research fields

While the use of qualitative case methods in technology transfer research is increasing, there has been no study of qualitative case methods on technology transfer research yet. In other fields of research, reviews on themes and trends of qualitative case methods are more common. For example, in the field of information system (IS) single and multiple qualitative case study approaches, according to Shakir (2002), is well suited to the field and is continuously growing in popularity amongst researchers. Similarly, Walsham (1995) notes the increase in studies in the IS field and highlights the challenges and implications for researchers pursuing this methodological approach. In the field of organisation studies, Locke and Golden-Biddle (1997) outline the use of qualitative case methods within the two highly regarded journals, namely *Academy of Management Journal* and *Administrative Science Quarterly*. They found that the published work of organizational studies scholars consists not only of the presentation of data and does not only array 'facts' and evidence logically but weaves persuasive practices into texts, even as the authors structure the coherence of the intertextual fields. Other fields also have noted the increased utilization of qualitative case methods such as management accounting (Scapens 1990), strategic management (Hoskisson et al. 1999), marketing (Easton 2010), international political economy (Odell 2001), SME firms (Chetty 1996; Perren and Ram 2004), logistics (Ellram 1996) and operations management (Meredith 1998). More recently in the field of entrepreneurship and small business, a study by Henry and Foss (2015) found a limited number of studies that use qualitative case methods and this had been applied across a variety of contexts such as family business, entrepreneurial networking and innovation. Henry and Foss (2015), based on their study, note that few papers published in entrepreneurship journals use qualitative case methods, suggesting that this research method is not fully accepted as a legitimate and rigorous approach within entrepreneurship scholarship and thus argue for a greater acceptance of the use of case methods.

2.2 Qualitative case methods and technology transfer research

The term 'case study' is strongly related to qualitative research designs, although it is utilized in many different ways (Ritchie et al. 2013). Case studies investigate a contemporary phenomenon where the boundaries between the object of study and context are unclear (Yin 2013).[1] Case study research is considered to be a preferred method of social science research in situations where the primary research questions are "how" and "why" questions, where the researcher is involved in doing an evaluation and where the researcher has minimal control over behavioural events (Bhattacherjee 2012). A qualitative case study approach involves the investigation of a contemporary phenomenon in its real-world context where the object of study or its environment is not manipulated. Easton (2010, p. 119) defines case research 'as a research method that involves investigating one or a small number of social entities or situations about which data are collected using multiple sources of data and developing a holistic description through an iterative research process'. This approach allows researchers to draw conclusions on the basis of a qualitative analysis from a single case study or from a comparative case study (Dul and Hak 2008). Woodside and Wilson (2003, p. 493) note that the principal objective of qualitative case study research is about 'achieving deep understanding of processes and other concept variables (e.g. actors' perceptions of their own thinking processes, intentions and contextual influences)'. In conducting qualitative case research, Stuart et al. (2002) break down case research into five stages: defining the research question, instruments development and site selection, data gathering, analysing data and disseminating the research findings.

Among scholars in technology transfer research, there is an increasing consensus that technology transfer processes and the corresponding technology transfer effectiveness are subject to contextual influences, i.e. the primary research question is "when" (Autio et al. 2014). Studies that had used qualitative case methods have highlighted the importance of context when studying different aspects of technology transfer. Some examples of this contextual importance focusing on the complexities involved include the qualitative case study of Jain and George (2007) of Wisconsin Alumni Foundation which highlights the need for TTOs to have dual purposes as well as the study of Kingsley, Bozeman and Coker (1996) on evaluating technology transfer from research, development and demonstration (RD&D) using 31 cases studies. As a result, research methods which take the importance of context into account and enable researchers to control for respective influencing factors gain in significance, yet there still remains a dominance of quantitative studies in technology transfer research.

2.3 Challenges and benefits of qualitative case methods research

Qualitative case methods research is not without its critics who originally viewed this method with scepticism arguing that they had unsystematic procedures and poor design and that they were less important than studies that produced generalisations for a whole population. Particular criticisms centred on internal and external validity (Stoecker 1991), on unclear conventions concerning case analysis (Miles 1979) as well as on rigor (Dubé and Paré 2003; Hoskisson et al. 1999). Specifically, Flyvbjerg (2006) identified and addressed what he examined as five misunderstandings of case study research including the theoretical knowledge, generalisability, generation of hypotheses, bias and appropriate

[1] We use terms in this section of the paper such case methods, case based research and case studies these all refer to qualitative case methods.

summaries of case studies. Moreover, Johnston et al. (1999) argue from a marketing domain perspective that case studies can contribute to theory building if researchers are systematic and rigorous in their approach. Furthermore, Welch et al. (2011) also argue that case studies can contribute to theory building. In the context of the international business field, they suggest that there are four methods of theorising in the field: inductive theory building, interpretive sensemaking, natural experiment and contextualised explanation. Applying case methods research in a rigorous manner can contribute to theory building and it needs to pass the 'tests of good theory' as Eisenhardt (1989) argues. Handfield and Melynk (1998) suggest and outline potential matching research strategies to theory building when using case based methods. For example, they suggest using unfocused in-depth case studies and longitudinal case studies when the purpose of the research is focused on discovery, yet where the research purpose is mapping then focused case studies, in-depth field studies multiple cases studies constitute more appropriate data collection approaches.

Case based research according to Darke et al. (1998, p. 275) can be used 'to provide descriptions phenomena, development theory and test theory'. However, Benbasat et al. (1987, p. 369) argue that 'case research is particularly appropriate for certain types of problems: those in which research and theory are at the early, formative, and, "sticky, practice based problems where the experiences of the actors are important and the context of action is critical"'. One of the real benefits of case methods affords researchers flexibility in approach while studying complex situations and contexts. Case methods research constitutes a useful research strategy for solving practical business problems and examining business theories but again represents a small percentage of total papers (Dul and Hak 2008). More recently, case methods have become scientifically rigorous and are considered by some researchers as a complementary method to the quantitative approach (Gagnon 2010). The validity of case study research can thereby be increased through triangulation with multiple means of data collection (Dul and Hak 2008).

3 Methodology

3.1 Systematic literature review

To examine the trends and themes of qualitative case methods research used in technology transfer research, we undertook a comprehensive systematic literature review of articles published in the following five academic journals: *Journal of Technology Transfer, Research Policy, Science and Public Policy, R&D Management* and *Technovation*. A systematic review was selected as opposed to the traditional narrative literature reviews. Traditional literature reviews have been broadly criticised for their inadequate relevance due to the utilisation of personal, subjective and biased methodology by authors (see Becheikh et al. 2006; Fink 1998; Hart 1998). Systematic literature reviews apply the same standards to secondary research as should be applied to primary research (Denyer and Neeley 2004). Attributes of the systematic review include the development of clear and precise objectives, pre-planned methods, a comprehensive search of all potentially relevant articles, the use of explicit criteria for article selection, an evaluation of the quality of the research and findings, an integration of the individual studies as well as an impartial and coherent delivery of the results (Transfield et al. 2003).

Systematic literature reviews follow an explicit, rigorous, and transparent methodology and therefore constitute a well-established approach in analysing trends and developments in a certain research field. In contrast to the traditional literature review, the systematic review aims to minimise error and bias in order to increase the quality of the review by using explicit and rigorous methods to identify, appraise and incorporate research on specific research questions (Ivarsson and Gorschek 2009). A further strength of the systematic literature review is its transparency and openness to critique (Pittaway and Cope 2007). The chosen systematic literature review is particularly useful for analysing large quantities of longitudinal data (Denyer and Neeley 2004).

The primary purpose of our systematic review is to identify key contributions to a specific field or research question, namely technology transfer research, and to descriptively present and discuss these findings. Given the growing body of research on technology transfer it is important to gain a common understanding of methodological approaches, respective terminology and research designs to consistently progress within in the field. Systematic literature reviews thereby involve two processes: the first process concerns defining review protocols and field mapping while the second process involves reporting the findings (see Transfield et al. 2003; Pittaway et al. 2004; Macpherson and Holt 2007).

3.2 Defining protocols

The protocol contains information on the precise questions that the study addresses, the study's population, the search strategy as well as the criteria for the inclusion and exclusion of studies (Transfield et al. 2003). This review's population was restricted to published academic articles in the following leading journals focusing on technology transfer research including the years from 1996 to 2015: *Journal of Technology Transfer, Research Policy, Science and Public Policy, R&D Management* as well as *Technovation*. A systematic Boolean keyword search was conducted within each journal including the terms: "Technology Transfer" AND ("Case Study" OR "Case Method" OR "Qualitative"). The studies that met the inclusion criteria specified in the protocol yielded 116 papers that used case methods utilizing the specific search strings. In order to validate the search results, we analysed the abstract of each of the 116 articles and checked for accuracy concerning our defined protocols, i.e. applied case methods in the context of technology transfer research. Upon removal of duplicate studies and papers that did not use case methods, such as special issue editorials, a total of 107 articles were included in the review (see "Appendix 1").

3.3 Mapping the field

Our final 107 articles were managed via a spreadsheet database using 23 fields (see "Appendix 2" for an excerpt of our spreadsheet database). The fields included the article's name, author(s) and their affiliations, year of publication, journal, duration of the study, theme, research question(s), keywords, location, unit of analysis, sector, number of cases, method(s), data collection, data analysis, data analysis software, findings and implications. These fields were chosen in order to allow for homogenous data extraction in preparation for the efficient analysis of the data.

3.4 Analysis

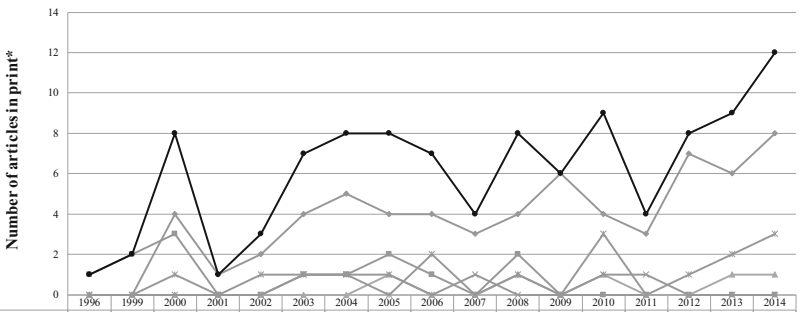
The process of data analysis is defined by Zikmund et al. (2013) as being the application of reasoning to understand and interpret the data that has been collected. The analysis of the data within the spreadsheet allowed for the identification and critique of the dominant themes that emerged from the systematic literature review as well as the key results and findings. Linking the themes across the various core contributions and subsequently highlighting such links is a necessary part of the analysis stage (Transfield et al. 2003). Reoccurring themes, units of analysis, sectors, data collection and data analysis methods were identified and recorded.

Our spreadsheet database served as a basis for the identification of commonalities among the 107 articles. By implementing a coding scheme and by in-depth analyses of technology transfer research trends, we were able to aggregate the papers and synthetize five distinct themes. We did this by each author coding individually and then we shared and agreed final coding to aggregate and synthetize five distinct themes. It is important to note that one article could thereby be assigned to more than one theme if appropriate. The following section outlines our key findings with respect to case methods trends and themes used in technology transfer research.

4 Findings

4.1 Range and number of articles

The total number of studies found within all five journals that met all the inclusion criteria specified in the protocol ranged from only one in 1996 up to twelve in 2014. This identifies a small increase in qualitative case methods studies over the period between 1996 and 2014. The individual journal that yielded the highest number of studies was the *Journal of Technology Transfer*, yielding 8 articles in 2014. In 1996, the journal *Research Policy* alone published one qualitative case methods study and not until 1999 were another 2 articles published; again, only in *Research Policy*. The fluctuating, minimal increase illustrated by the line graph (see Fig. 1) identifies that although there has been a slight



	1996	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
Journal of Technology Transfer	0	0	4	1	2	4	5	4	4	3	4	6	4	3	7	6	8
Research Policy	1	2	3	0	0	1	1	2	1	0	2	0	0	0	0	0	0
Science and Public Policy	0	0	0	0	0	0	0	1	0	0	1	0	1	0	0	1	1
R&D Management	0	0	0	0	0	1	1	1	0	1	0	0	1	1	0	0	0
Technovation	0	0	1	0	1	1	1	0	2	0	1	0	3	0	1	2	3
Total Number of Articles	1	2	8	1	3	7	8	8	7	4	8	6	9	4	8	9	12

Fig. 1 Number of articles in print per year and per Journal. *Asterisk* The year 2015 has been excluded from the diagram as our data collection did not include the full year 2015, but only the first three months of 2015

increase in qualitative case methods studies, it is still an under-utilized approach. The total number of articles published in 2015 gives rise to only 2 articles. As the systematic literature review was undertaken in the first quarter of 2015, some articles of 2015 had not yet been published giving rise to the low number of articles in this year. In general, the number of published articles that have used case methods is comparable low, suggesting that other methodological approaches are more dominant.

4.2 Main research themes

From our individual coding and through the authors' collective consensus, we identified five thematic categories within the five journals between 1996 and 2015 (see Fig. 2). We have detailed keywords for each of our five themes (see Fig. 2) and based on this, there is a huge variety of keywords. We also conducted a similar exercise for our sectoral analysis. The most common reoccurring theme is 'Technology Transfer Mechanisms and Technology Transfer Offices' giving rise to a total of 70 articles. The *Journal of Technology Transfer* has published the largest number of articles related to this theme category with a frequency of 37 articles, followed by the journal *Technovation* with a frequency of 16. The next most common reoccurring theme is 'R&D and firm knowledge transfer' giving rise to a total of 56 articles with *Journal of Technology Transfer* publishing more than half of this total. The least common reoccurring theme is the thematic category of 'University-Industry Collaboration' with a total theme occurrence of 15. *R&D Management* proved to be the journal with the least number of reoccurring themes accumulating a total of 14 occurrences of the thematic categories. *Research Policy* in turn provided for a total of 29 of the thematic categories. The thematic category of 'Academic Entrepreneurship' produces a frequency of 46 occurrences with the *Journal of Technology Transfer* accounting for 63 % of total occurrences. The thematic category of 'Commercialization' accounted for 17 % of the total thematic categories, again with *Journal of Technology Transfer* accounting for the most occurrences, followed by *Technovation*. The dominant focus on technology transfer mechanisms and TTOs reflects a predominant focus within the field using other methodological approaches.

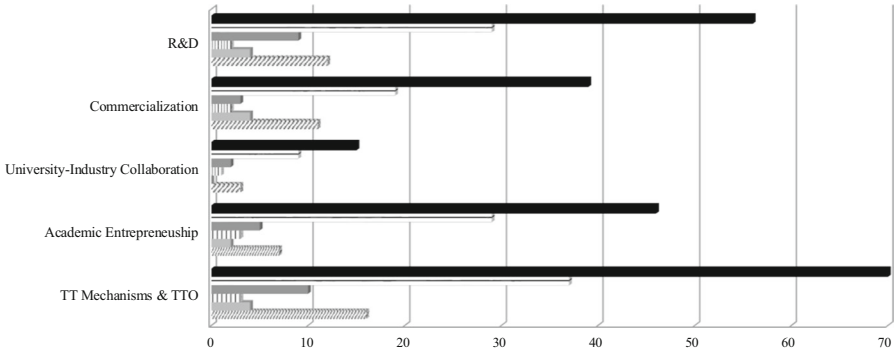
4.3 Authorship

Using qualitative case methods research data collection can take longer and involve multiple sources. Table 1 illustrates the number of authors who have published more than one article related to case methods in technology transfer research. No author between 1996 and 2014 has published more than 3 papers, highlighting the lack of research and revisited research within the field of technology transfer research. Similar results have been derived by Ghio et al. (2015) in their in-depth co-authorship analysis focusing on the evolution of the 'Knowledge Spillover Theory of Entrepreneurship'. Their study revealed the concentration of this strand of literature on a limited set of central authors.

Most of the 107 case methods studies have been published by a single author accounting for 34 studies, followed by 31 dual authorships and 29 articles written by three authors. There are 11 studies that were written by 4 authors and only 2 papers written by 5 authors (see Fig. 3).

4.4 Location of paper and affiliation of author

Qualitative case studies are always associated with certain contexts. Interestingly, some contexts triggered more in-depth analyses than others, leading to the question which



	TT Mechanisms & TTO	Academic Entrepreneuship	University-Industry Collaboration	Commercialization	R&D
Total Themes	70	46	15	39	56
Journal of Technology Transfer	37	29	9	19	29
Research Policy	10	5	2	3	9
Science and Public Policy	3	3	1	2	2
R&D Management	4	2	0	4	4
Technovation	16	7	3	11	12

Fig. 2 Theme analysis. *Note*. Five different thematic categories were chosen allowing for thematic keywords from each paper to be grouped under their appropriate heading. **Technology Transfer and Technology Transfer Offices includes (excerpt):** Technology transfer, Technology transfer offices, Knowledge transfer, Technology transfer consortium, Technology transfer performance index, Technology diffusion, Technology transference, Transfer Mode, Technology life cycle management, Knowledge management system (KMS), Boundary spanner/spanning, Academic patents, Absorptive capacities, Technological Capability. **Academic Entrepreneurship includes (excerpt):** University entrepreneurial ecosystems, Entrepreneurship education, Entrepreneurial universities, University entrepreneurship, Academic entrepreneurship, Entrepreneurship, Entrepreneurial scientists, Opportunity entrepreneurship, Necessity entrepreneurship, Higher education, University spinoffs and start-ups, Academia, Communication network, Academic spin-off, Start-up, Spin-off, Spin-out, Individual motivations, University incubators, Proof of concept (POC). **University-Industry Collaboration includes (excerpt):** Technology development stages, Networks, University-industry linkage, University–industry collaboration, University–industry research partnerships, Strategic alliance, University–industry relationships, Dynamics of collaboration, Process studies of collaboration, Network formation, Relationship management. **Commercialisation includes (excerpt):** University technology commercialization, Technology commercialization, Commercialization, Scientific-knowledge commercialisation, IP commercialization, External patent exploitation, Intellectual property, Licensing, Desorptive capacity, Case studies. **R&D includes (excerpt):** Technology evolution, Innovation, Innovation management, Management of innovation, Innovation policy, Innovation synergy, Product innovation, Eco-innovation, Fuzzy front end of innovation, Total innovation management, Construction, Knowledge flows, Firms' internationalization, Paradigm change, Biotech, Valley of death, Public/private collaboration, Inter-organizational relations, R&D management, Industrial R&D, R&D, R&D researcher, Internationalisation of R&D, R&D strategy, Market-driven R&D, Technology-driven R&D, Cost and organization of R&D, Government research and development, Cross-sector R&D, R&D cooperation, Breakthrough technologies, Agency theory, Research organization, Research policy, Research centres, Research productivity, Research centre, Stem cell research, Cooperative Research Centres, Organizational change, Funding strategies, Scientific publications, Multinational enterprises, Corporate venturing, Corporate strategy, Program evaluation, Benefit-cost analysis, Internal rate of return, Net present value, Green nanotechnology, Evolutionary economics, Evolutionary capabilities

contexts are mostly studied and why. In examining the location of the paper and affiliation of the authors, we were interested in examining the country location of the case methods paper and whether authors were utilising data from their own country of affiliation or other countries. Figure 4 outlines which countries each of the case methods papers are based on as well as the country of affiliation of the author. Out of the total 107 papers, 28 of these are based on the USA, followed by 16 studies on the UK and 7 studies on Germany, Italy

Table 1 Authors with 2 or more papers

Author	No. papers
Gassmann, Oliver	2
Gibson, David V.	2
Guerrero, Maribel	2
van Pottelsberghe de la Potterie, Bruno	2
Xu, Qingrui	2
Gulbrandsen, Magnus	3
Link, Albert N.	3
Lockett, Andy	3
Siegel, Donald S.	3
Wright, Mike	3

and China. Some 60 authors are affiliated with the USA, followed by 29 authors from the UK, 18 from Spain, 18 from Canada and 12 from Italy. Some 45 authors affiliated with the USA wrote papers about the USA indicating a 75 % rate of authors affiliated with the USA who wrote papers about their own country.[2] Some 20 authors affiliated with the UK have completed studies on the UK giving rise to a 68 % rate. The 12 Spanish authors completed studies based on Spain giving a 67 % rate and similarly 12 Canadian authors completed studies on Canada indicating a 67 % rate and indicating a consistently high level of sample convenience. These results indicate that the location of the authors matters: authors utilize their personal networks and relationships to conduct case methods research.

A minority of authors completed studies not based on their affiliated country. Some 9 authors from the USA wrote papers that were not based on the USA indicating a rate of only 15 %, while 7 authors from the UK completed studies that were not based on the UK indicating a rate of 24 %. Some 80 % of French authors however, have completed studies based on countries other than France while all 3 of the Danish authors completed their studies on foreign countries. Some 5 USA affiliated authors have completed studies that are based on both the USA and other countries while 4 Swiss and 4 Irish authors have completed studies on both their home country and foreign countries. These findings would suggest that researchers' dominant preferences are to focus their studies in the country where they are affiliated to and there are small numbers of authors who base their studies other countries. Furthermore, our findings show that there is a US dominance for location of papers and affiliation of authors and there is an under-representation of emerging and Asian economies as well as from countries where economies are well advanced in terms of R&D investment and innovation such as Finland and the Netherlands.

4.5 Sectoral contexts

Seven sectoral categories were identified from the analysis of the 107 qualitative case methods studies based on individual author coding and through the authors' collective consensus. The 'manufacturing industry' yielded a total of 66 occurrences, within *Journal of Technology Transfer* accounting for almost half of these, followed by *Technovation* accounting for 22 articles. Although the *Journal of Technology Transfer* yielded the largest overall occurrences of the 'manufacturing sector' almost half (46 %) of all studies within

[2] Due to multiple authorships, the number of authors exceeds the number of papers.

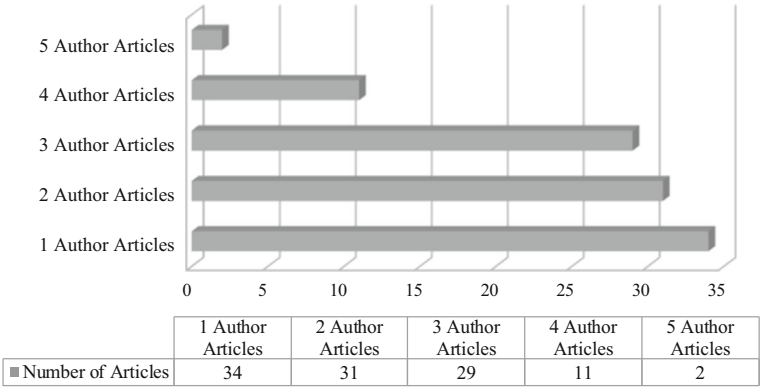


Fig. 3 Number of authors per article

the journal *R&D Management* had a focus on this sector. The next most analysed sector was 'health and biotechnology' accounting for a total of 53 occurrences, again with *Journal of Technology Transfer* accounting for 62 % of all instances of this sector. The 'software and ICT' sector yielded a total of 49 occurrences with *Journal of Technology Transfer* accounting for half of the total occurrences. The sector with the least amount of focus was the 'chemical industry' yielding a total of only 7 occurrences. The journals *Research Policy*, *Science and Public Policy* as well as *Technovation* yielded no instances of the 'chemical industry'. Overall our findings highlight the significant range of sectoral contexts that have been used by authors using case methods research in the field of technology transfer. This range of sectoral contexts has yet changed its focus over time. Whereas the manufacturing industry had previously been the main focus of case methods studies, this focus has now changed towards high-tech and emerging industries such as the health and biotechnology, the software and ICT as well as the energy and renewable resources industry. One explanation for this general development towards knowledge-intense industries is provided by Audretsch et al. (2014) as they argue that the emergence of new technologies functions as the driving force and enabling factor to globalization (Fig. 5).

4.6 Data collection

We categorised the primary and secondary data collection methods based on the descriptions used by the authors in their published papers (see Fig. 6). We found that the most frequently used primary data collection methods were semi-structured face-to-face interviews, telephone interviews and questionnaires which we subsumed under the category 'interviews'. Furthermore, we identified nine descriptors used by authors in outlining primary data collection using interviews. Such methods of primary data collection would be considered a traditional one among qualitative researchers. Other means of collecting primary data collection such as focus groups and action research have been used less frequently, indicating that authors are favouring more traditional and safe approaches to data collection. Interestingly, authors used a wide variety of descriptions in relation to primary data collection, particularly those associated with interviews. Consequently there is a need for future researchers to be clearer and more precise use of terminology to describe primary data collection.

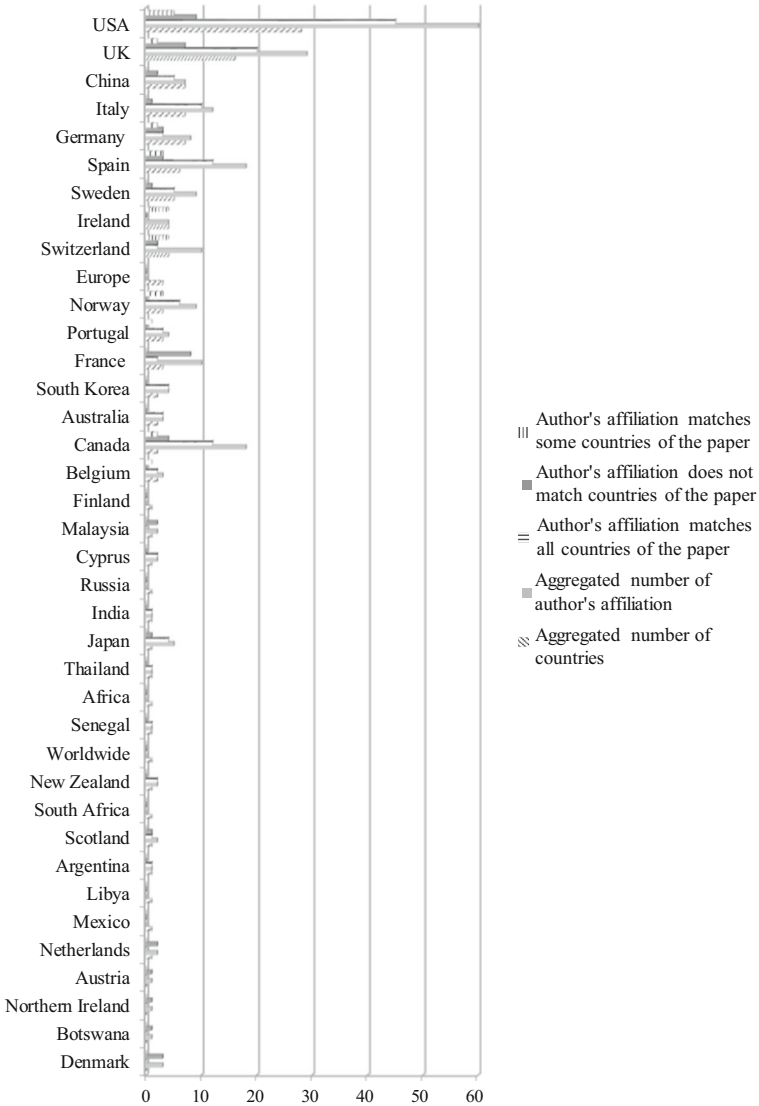
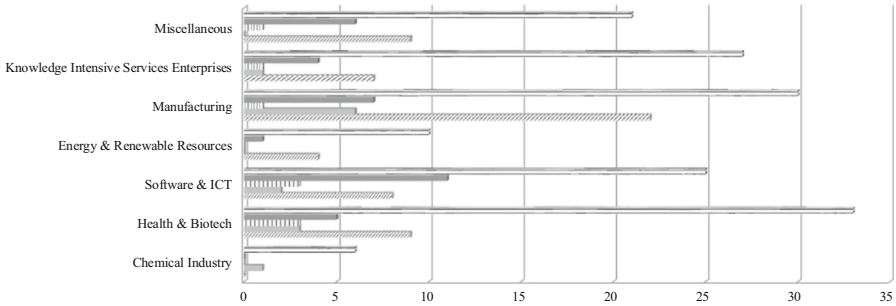


Fig. 4 Location of papers. *Note*. (1) When the paper's country is 'Worldwide', an author e.g. from the USA is treated as if writing about a country which is not affiliated country. (2) When the paper's country is 'Europe', an author from a country within Europe e.g. Germany is treated as if writing about a country which is not affiliated country. (3) An author from a country within the UK e.g. Northern Ireland writing about the UK is not considered an affiliate of the UK. (4) When a paper does not mention which country the paper is based on, no country is assigned

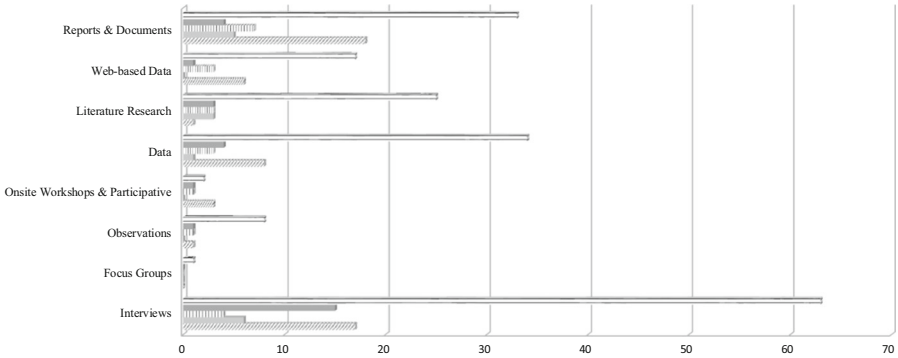
The top three descriptors we found in analysing the papers in our study for secondary data collection methods have been data reports, websites and newspapers. What is interesting to note is the wide variety of secondary data collection sources that we found in analysing the papers in our study and the frequency of data from previous studies, visual records and archival records. These findings suggest that authors using qualitative case



	Chemical Industry	Health & Biotech	Software & ICT	Energy & Renewable Resources	Manufacturing	Knowledge Intensive Services Enterprises	Miscellaneous
Journal of Technology Transfer	6	33	25	10	30	27	21
Research Policy	0	5	11	1	7	4	6
Science and Public Policy	0	3	3	0	1	1	1
R&D Management	1	3	2	0	6	1	0
Technovation	0	9	8	4	22	7	9

Fig. 5 Sectoral Analysis. *Note.* **Chemical industry includes:** Chemical–Pharmaceutical, Chemicals, Chemical and materials technology, Chemicals and pharmaceuticals, Materials-chemistry, Agrochemical. **Health and Biotech include:** Assistive Technology, Biochemistry, Bioengineering, Biology, Biomedical research, Biological, BioImaging, Biomass power equipment, Biomedical, Biomedicine, Bioscience, Biotechnologies, Biotechnology, Chemical medicine, Chemicals-cosmetics-medical equipment, Health, Instrumentation and Medical Pharmaceutical, Instrumentation and measurement technology, Medical, Medical research, Medical technology, Neuroscience, Pharmaceuticals, Pharmacy, Protein science, Rehabilitation Engineering Research, Science, Science and Engineering, Science and technology sector, Life Science, Stem cell research. **Software and ICT include:** Information Technologies, Communications, Computer Hardware, Internet, Software, High technology, ICT, IT, ITSIS, Informatics and telecommunication, Instrumentation and laser systems, Localisation, Microelectronics technology, Software, Software Piracy, Software design and development, Systems integration services, Technology, Telecommunications. **Energy and Renewable Resources include:** Energy, Environment, Environmental, Environmentally sound technologies, Pollution prevention industry, Environmental protection, Oil and gas. **Manufacturing includes:** Aerodynamics, Aeroengine, Aeronautics, Aerospace, Aluminium industry, Auto industry, Automotive, New energy automobile industry, Nanotechnology, Cable Production, Semiconductors, Computer-integrated manufacturing, Construction, Electronic goods manufacturing, Electrical construction, Engineering and physical sciences, Engineering technologies, Furniture manufacturing, Iron and Steel Industry, Manufacturing, Materials technology, Mechanical Engineering, Mechanics-electronics, Metals, Microelectronic Manufacturing, New materials, Plastic and rubber, Plastics industry, Production technology, Production, Robotics, Heavy-duty robotics, Semi-conductor, Ship building, Solar Cell industry, Textile, Traditional Sector, Vehicle catalyser manufacturing, Material Science. **Knowledge Intensive Services Enterprises include:** Consultancy, Education, Financial services, Government engineering research, Industrial-oriented research, Knowledge intensive business services, Mathematics, National interest research, Physics, R&D testing services, R&D into configuration and functioning of asphalt plants, R&D into dynamics of turbine machine, R&D into intravolume marking laser system, R&D into labelling machine, R&D into materials for actuators, R&D into measurement system for vehicle steadiness, R&D-intensive FDI, Services, Research Sector. **Miscellaneous includes:** Architecture, Charity, Creative sector, Dairy sector, Design and fashion, Dual-use Science & technologies, Electronics, Optoelectronics, Entomology, Mechanic and architectural sectors, Food and agriculture, Food and nutritional science, Functional foods, Foodstuff, Forest science, Horticulture, Hotels and restaurants, Industrial sectors, Media, Military (Air Force), Oceanography, Photonics, Plant physiology, Railroad Industry, Retail and wholesale, Space sector, Tourism, Transport, Water Purification, Building

methods are using more multiple sources of secondary data. From the descriptions of primary and secondary data collection in our study that authors use a variety of terms such as for data. Our findings suggest that authors in using qualitative case methods are opting for more traditional methods of data collection and we suggest that one of the drivers of

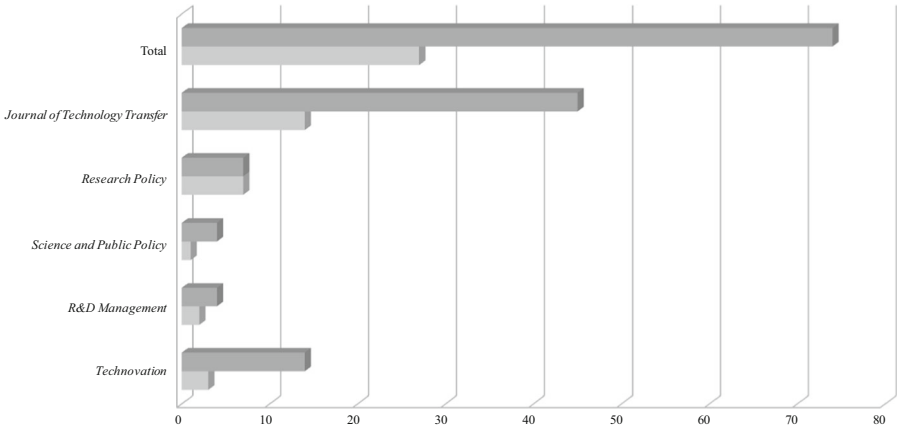


	Interviews	Focus Groups	Observations	Onsite Workshops & Participative	Data	Literature Research	Web-based Data	Reports & Documents
Journal of Technology Transfer	63	1	8	2	34	25	17	33
Research Policy	15	0	1	1	4	3	1	4
Science and Public Policy	4	0	1	1	3	3	3	7
R&D Management	6	0	0	0	1	3	0	5
Technovation	17	0	1	3	8	1	6	18

Fig. 6 Primary and secondary data collection analysis. *Note.* Data collection methods were categorised into primary data collection and secondary collection. Four categories were then selected for primary data collection and four for secondary data collection. The *Primary Data Collection* data collection categories are Interviews, Focus Groups, Observations and Onsite Workshops and Participative. **Interviews include:** Closed question interviews, Face-to-face interviews, Informal, Conversational face-to-face interviews, Open unstructured question face-to-face interviews, Open-ended question interviews, Open-ended semi-structured interviews, Semi-structured face-to-face interviews, Semi-structured questionnaire, Structured interviews, Telephone interviews, Unspecified Interviews, Web-based/Postal Questionnaire, Voice recordings. **Focus groups include:** Focus groups. **Observations include:** Observations. **Onsite Workshops and Participative include:** Meetings, Onsite visits, Participative action research, Workshops. The *Secondary Data Collection* data collection categories are Data, Literature Review, Web-Based Data and Reports & Documents. **Data includes:** Academic articles, Conference data, Data from previous studies, Database data, Employment data, Experimentations, Financial data, First-person accounts, Industry data, IP-related law, Published data, TTO data, Unpublished data, Websites/web-based data, Biographical and autobiographical materials. **Literature Review includes:** Literature research. **Web-based Data includes:** Websites. **Reports and Documents include:** Business Plans, Manuals, Newspapers, magazines, newsletters, journals, brochures, Presentations, Press review, Publicly accessible policies, Reports and Documentations, Charts, Visual records, Archival records, Experts and market information

this is around ensuring that their studies have a higher probability of being published and convincing reviewers of the rigor that they have applied to their case methods.

One of the questions facing case methods researchers is how many data sources are required to conduct a study. We found from the papers in our study that frequency of multiple data collection was higher to single by nearly a 3 to 1 ratio (see Fig. 7). Due to the multitude of perspectives derived from multiple data collection methods, the robustness of research results increases by having more than just one data source. Consequently, multiple data collection methods increasingly gain in importance (Creswell 2013). The qualitative case methods articles published in *Research Policy* utilise an equal number of single data collection methods to multiple data collection methods. This clarifies the wider use of mixed-modes of data collection in qualitative case methods studies. Our findings would suggest that data authors in designing case methods papers should select their data collection approaches carefully and consider a data collection approach that fits their research question and helps them to support their argumentation most effectively.

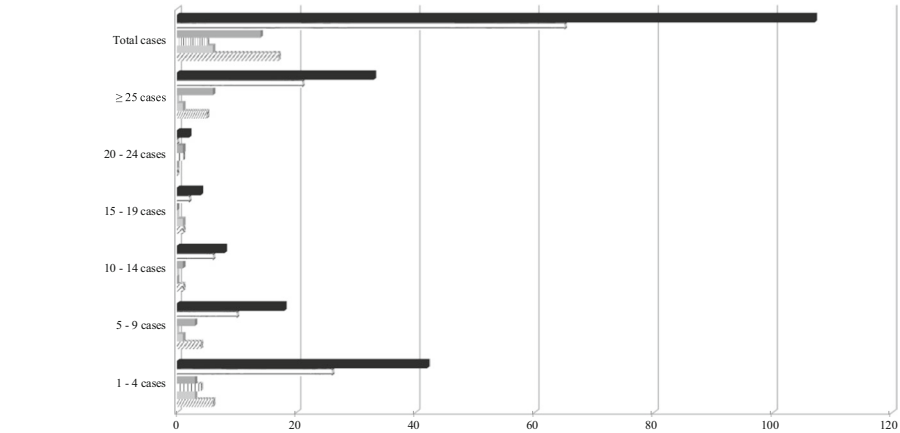


	Technovation	R&D Management	Science and Public Policy	Research Policy	Journal of Technology Transfer	Total
Multiple Data Collection	14	4	4	7	45	74
Single Data Collection	3	2	1	7	14	27

Fig. 7 Data collection approach

4.7 Number of cases

A key issue for qualitative case methods researchers is how many cases are needed for their studies. In analysing the papers, we found that 42 papers used between one and four cases and 33 papers used more than 25 cases (see Fig. 8). This suggests that there are two case extremes emerging, those studies that rely on a small number of cases and authors that used a significant number of cases for their studies. We suggest that this heterogeneity in the number of cases is beneficial for the field of technology transfer research as it gives case



	1 - 4 cases	5 - 9 cases	10 - 14 cases	15 - 19 cases	20 - 24 cases	≥ 25 cases	Total cases
Total	42	18	8	4	2	33	107
Journal of Technology Transfer	26	10	6	2	0	21	65
Research Policy	3	3	1	0	1	6	14
Science and Public Policy	4	0	0	0	1	0	5
R&D Management	3	1	0	1	0	1	6
Technovation	6	4	1	1	0	5	17

Fig. 8 Number of cases

methods researchers continued flexibility as to the number of cases as our findings clearly show that journals in our study publish papers using both low and higher number of cases.

4.8 Data analysis and data analysis software

There is a growing array of data analysis software available to support qualitative case methods researchers in analysing data. We found that studies utilizing data analysis software constitute a minority. Only 14 papers used data analysis software and the most used analytical software tool was N-Vivo analysis (see Fig. 9). The N-Vivo analysis software was used three times in 2014 and once in 2010 by articles published in the *Journal of Technology Transfer* and *Technovation.*

In analysing the methods of data analyses we found that authors used a wide variety of descriptions in relation to the data analysis they used (see Fig. 10). These descriptions were grouped under five data analysis categories. Some 42 % of the 107 studies used a form of case study analysis while 22 % used a method of qualitative analysis, some 14 % used data triangulation, 14 % for miscellaneous and less than 10 % used literature analysis.

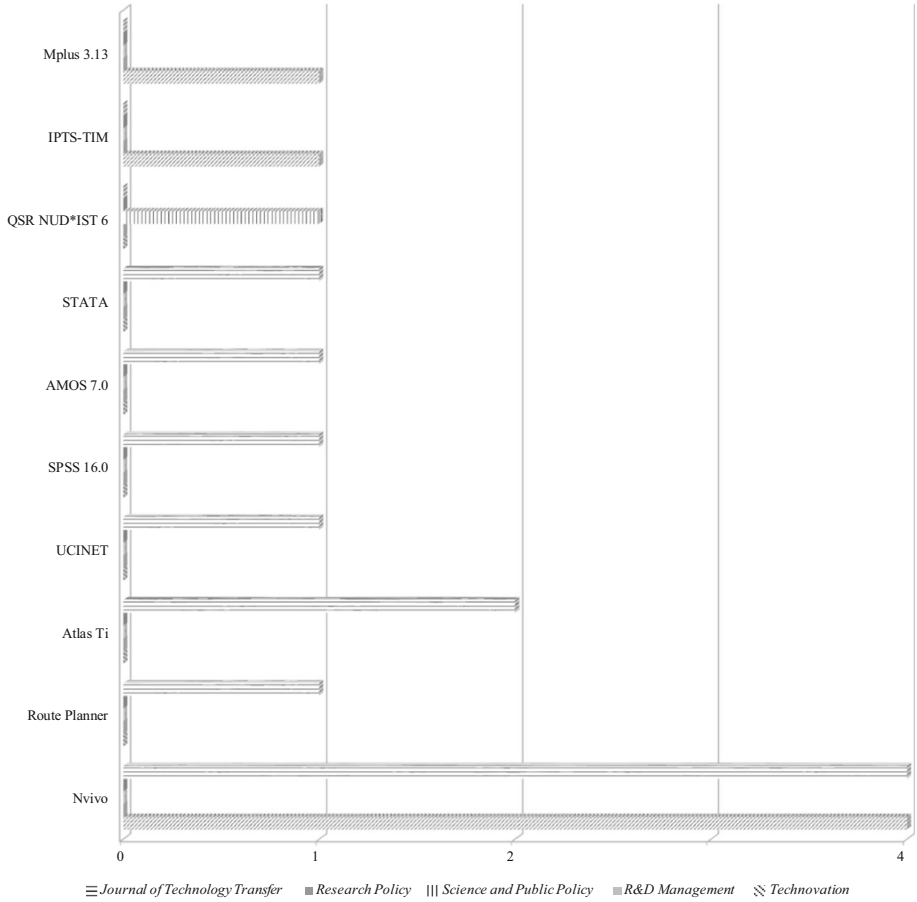
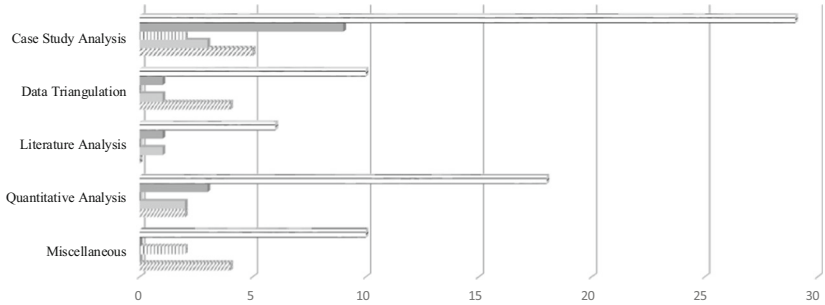


Fig. 9 Data analysis software



	Miscellaneous	Quantitative Analysis	Literature Analysis	Data Triangulation	Case Study Analysis
Journal of Technology Transfer	10	18	6	10	29
Research Policy	0	3	1	1	9
Science and Public Policy	2	0	0	0	2
R&D Management	0	2	1	1	3
Technovation	4	2	0	4	5

Fig. 10 Data analysis. *Note*. Methods of data analysis were grouped under 5 categories. **Case Study Analysis include:** Case study analysis, Qualitative data analysis, Cross-case analysis, Multiple case study analysis, Within-case analysis, Content analysis, Longitudinal multiple case studies, Longitudinal comparative case study, Multiple case study coding, Observation and document analysis, Case by case pattern of analysis. **Data Triangulation include:** Data triangulation. **Literature Analysis include:** Literature analysis. **Quantitative Analysis include:** Factor analysis, Benefit-cost analysis, Data envelopment analysis, Exploratory factor analysis, Hierarchical regression, Patent data analysis, Regression analysis, Survey analysis, Axial coding, Coding scheme, Codification of Questionnaires, Exploratory analysis, Descriptive analysis/Statistics, Data categorization, Cluster analysis. **Miscellaneous include:** Critical analysis, Network analysis, Cross-cultural analysis, Cross-sectional analysis, Ethnographic analysis, Explorative analysis, Failure mode effect analysis, Functional analysis, Grounded theory approach, Synthesis of existing literature, Thematic analysis, Between Group Analysis, Center-level analysis, Comparative analysis, Comprehensive mapping, Concept analysis

5 Discussion

What is overall evident from our study is that qualitative case methods are still in an emergent state within the field of technology transfer research. Thus, continuing to progress qualitative case methods based studies have an important contribution to make to advancing knowledge and for theory building. It also places onus on case methods researchers to be conscious in the description and precision in how they outline case methods used, to provide more detailed accounts of the data collection and analysis they used as well as being more experimental in using a wider range of data collection methods and analytical approaches in further studies. Moreover, our findings highlight that top journals do publish qualitative case methods research from a variety of sectoral contexts using single and multiple data collection sources.

In reflecting on the trends with respect to affiliated locations of authors, our findings especially concerning North America, UK and Canada, which is not surprising given the emergence and growth of the field of technology transfer research from North America in particular over the last three decades. Political interventions such as the Bayh-Dole Act in North America (Mowery et al. 2015) triggered qualitative case methods research focusing on the effects of technology transfer. What is surprising is that there is a growing spread of affiliated locations throughout different continents. However, there is a deficit with respect to emerging continents such as Africa, emerging Eastern European countries and high growth economies such as India. For countries with advanced national innovation systems

(Castellacci and Natera 2013) such as Japan and South Korea it is surprising that we did not find more qualitative case methods studies. More studies from these regions are needed that can both offer further contributions to developing the field but also insights into different contextual approaches and practices.

The predominant research theme on technology transfer mechanisms and TTOs that we found is not surprising as it forms the basis for all other strands of literature within technology transfer research (Villani et al. 2016). In aggregating this category for the purposes of our study it proved challenging to condense different articles due to the growth of new terms such to describe technology transfer as 'knowledge transfer', 'technology transferences', etc. The same issues occurred within the research field of academic entrepreneurship and how it is interpreted as a term within the field. Behind our findings, it gives rise to an interesting issue, namely that of interpretations of commonly used terms within the field. On the one hand loose and growing terms to describe a phenomenon under study can be helpful in growing and developing a field. On the other hand, it can also create confusion and potentially undermine the empirical results and intentionality of contributions from researchers. Particularly with respect to qualitative case methods research, researchers have to be particularly conscious and consistent with their use and interpretation of key themes that they build their studies around. At the current stage of the development of the field of technology transfer research, there is a need to reflect on commonly used terminology and its utilization within the field as well as how it should be understood for future studies. The consistent and precise utilization of terminology thereby has to comprise both content and methodology to overcome critics designating case methods as less rigorous (Dubé and Paré 2003; Hoskisson et al. 1999). Moreover, qualitative case study researchers need to be aware of the criticisms of using this approach particularly in relation to validity (Stoecker 1991), formulation of case analysis (Miles 1979), as well as bias, generalisability and hypotheses generation (Flyvbjerg 2006). In designing studies using qualitative case methods, researchers need to consider carefully these criticisms, also with regard to their intended contribution, i.e. choosing the appropriate data collection approach. Rigor has to pervade all aspects of qualitative case methods researcher activities. This for example requires more explicit descriptions of data analysis conducted such as first and second order coding that was undertaken by the (single or multiple) case analysis (see O'Kane et al. 2015a, b). Given the movement of the field toward a micro level analysis embracing and utilising more data collection approaches and data analysis, techniques can make a substantive contribution to generating new knowledge as well as providing insights that are relevant to technology transfer practitioners (Cunningham et al. 2016). It is clear that there is much scope for more case methods research developing within the field of technology transfer and within the broad themes that we have identified.

Our findings show that there is a focus on a limited number of data collection approaches with interviews, data, reports and documents being the predominate sources of data. Within the interview category, we found a wide variety of descriptions used. For data, reports and documents researchers used a wide variety of sources. It is encouraging to find such an array of sources. However, we suggest that there is room for the use of more data collections sources that would further enhance qualitative case methods studies. Approaches such as ethnographic methods, participant observations and focus groups can contribute to the development of the field. We suggest that there is a need for even more plurality of data collection methods as this will deepen and strengthen further the field of technology transfer research as well as complement more quantitative studies. Also in furthering rigor, there are significant opportunities for qualitative case methods researchers

to adopt data triangulation, which is used by only a small percentage of researchers in our study.

For qualitative case methods researchers, our study shows that there is a nearly 3 to 1 ratio of multiple data collection to single data collection. This would suggest that multiple data collection is favoured more by researchers and also by the journals in our study but also is reassuring that single data collection approaches are being published by high impact journals in the field. This has implications for qualitative case methods researchers in terms of theory development and generalizability of their studies. Our study has found that there is a range of data collection methods and the implications for researchers using qualitative case methods is that they have to be clear what contribution they are seeking to make which in turn determines the range for their study. Where we see further opportunities is in relation to growing multiple data collection across countries.

The question how many cases are sufficient for a qualitative case methods study is one that is of concern for qualitative case methods researchers. Our findings show that the majority of researchers used less than four cases or more that twenty-five cases. Within the field we are seeing two extremes and our breakdown of the journals in our study would indicate that journals are open to publishing papers with a range of number of cases. Small numbers of cases allow for in-depth analyses. Larger numbers of cases might support the generalizability of the findings. What surprised to us is the limited use of data analysis software. As the field evolves further, there is a need for case methods researchers to be more descriptive in relation to the analysis they conducted irrespective of whether they used data analysis software or not. It would further increase the efficiency of data preparation and exploitation in case of large amounts of data.

Finally, our analysis revealed a very broad sectoral activity focus with an emphasis on certain sectors such as 'health and biotechnology' as well as 'manufacturing'. We suggest that there is a need to both focus on sectors that are constantly increasing in importance such as the energy industry shifting towards renewable energies or knowledge intense service enterprises as well as on cross-sectoral studies combining different perspectives from different industries. Thus, scholars should establish a more holistic picture concerning technology transfer processes and corresponding effectiveness in distinct sectors. Also there are opportunities to apply qualitative case methods to examining technology transfer processes in the digital and circular economies. Overall, opportunities also lie in advancing knowledge in the field using longitudinal sector data that utilizes qualitative case methods and quantitative approaches.

6 Conclusion

We believe that our extensive systematic literature review on qualitative case methods in technology transfer research has produced three important implications for the field. First, future qualitative case methods researchers should reflect on used terminology and its utilization both concerning content and methodology. Only a consistent and adequate utilization of terminology enables the advancement of technology transfer and qualitative case methods research to address robustly and overcome on-going criticisms of case methods. To address this, case methods researchers should be clear in their use of terminology, research objectives, data collection and data analysis. A systematic approach to rigor by researchers allays critics, retains the flexibility of approach that is required and also contributes in an effective way to the on-going development of the field.

Second, there is a clear need for more plurality of data collection methods in combination with a more intense usage of data analysis software. On the one hand, this enables researchers to advance the field of technology transfer research as well as to complement more quantitative studies. On the other hand, data analysis software allows for processing larger amount of data, resulting in the possibility of more in-depth analyses. We would encourage qualitative case methods researchers to be more ambitious and innovative in their data collection approaches. This boldness and ambition is necessary to develop the field around the five themes we have identified, but also in opening up new research themes.

Third, case methods in technology transfer research should make use of cross-sectoral as well as cross-country research designs to again allow for more in-depth analyses as well as to make use of different perspectives to create a more holistic picture concerning the field of technology transfer research. The combination of both qualitative and quantitative research designs opens up new research trajectories to create further insights into the processes and mechanisms of technology transfer as well as its corresponding effectiveness.

In conclusion, Flyvbjerg (2006) argues that a 'scientific discipline without a large number of thoroughly executed cases studies is a discipline without systematic production of exemplars, and a discipline without exemplars is an ineffective one'. Our paper highlights essential necessities and opportunities for qualitative case methods researchers to further contribute to the advancement of knowledge in the field of technology transfer research and qualitative case methods in general.

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Compliance with ethical standards

Conflict of interest The authors have no conflict of interest.

Appendix 1

See Table 2.

Table 2 Overview of themes and articles

Author(s)	Journal
Theme: technology transfer mechanisms and TTOs	
Acworth (2008)	Research Policy
Albors et al. (2006)	Journal of Technology Transfer
Bach et al. (2002)	Journal of Technology Transfer
Backman et al. (2007)	R&D Management
Bathelt et al. (2010)	Technovation
Becker and Gassmann (2006)	Journal of Technology Transfer
Bessant (1999)	Research Policy
Bianchi et al. (2011)	R&D Management

Table 2 continued

Author(s)	Journal
Boehm and Hogan (2014)	Journal of Technology Transfer
Cacciatori (2008)	Research Policy
Chapple et al. (2005)	Research Policy
Daellenbach and Davenport (2004)	Journal of Technology Transfer
Derrick (2014)	Journal of Technology Transfer
Díaz (2012)	Journal of Technology Transfer
Doganova (2013)	Science and Public Policy
Efstathiades et al. (2000)	Technovation
Fako et al. (2000)	Journal of Technology Transfer
Festel (2013)	Journal of Technology Transfer
Giaretta (2013)	Journal of Technology Transfer
Golob (2006)	Journal of Technology Transfer
Greiner and Franza (2003)	Journal of Technology Transfer
Guerrero and Urbano (2012)	Journal of Technology Transfer
Hansen and Ockwell (2014)	Technovation
Hess and Siegwart (2013)	Journal of Technology Transfer
Howells (2006)	Research Policy
Hussler et al. (2010)	Technovation
Intarakumnerd and Charoenporn (2015)	Research Policy
Jackson and Audretsch (2004)	Journal of Technology Transfer
Jolly (2004)	R&D Management
Juanola-Feliu et al. (2012)	Technovation
Kingsley et al. (1996)	Research Policy
Knie and Lengwiler (2008)	Science and Public Policy
Kroll and Liefner (2008)	Technovation
Kumar et al. (2007)	Journal of Technology Transfer
Landry et al. (2013)	Technovation
Leahy (2003)	Journal of Technology Transfer
Leitch and Harrison (2005)	R&D Management
Link and Siegel (2005)	Research Policy
Maia and Claro (2013)	Journal of Technology Transfer
Malik (2002)	Technovation
Marcotte and Niosi (2000)	Journal of Technology Transfer
Markman et al. (2004)	Journal of Technology Transfer
Martinelli et al. (2008)	Journal of Technology Transfer
Matsumoto et al. (2010)	R&D Management
Moncada-Paternò-Castello et al. (2003)	Technovation
Moore and Thiongane (2000)	Journal of Technology Transfer
Morandi (2013)	Journal of Technology Transfer
Moultrie (2015)	Technovation
Nilsson et al. (2010)	Journal of Technology Transfer

Table 2 continued

Author(s)	Journal
Owen-Smith and Powell (2001)	Journal of Technology Transfer
Park et al. (2010)	Journal of Technology Transfer
Pittiglio et al. (2009)	Journal of Technology Transfer
Rasmussen et al. (2006)	Technovation
Saragossi and de la Potterie (2003)	Technovation
Resende et al. (2013)	Technovation
Salvador (2011)	Journal of Technology Transfer
Sanders and Miller (2010)	Science and Public Policy
Saragossi and de la Potterie (2003)	Journal of Technology Transfer
Schoen et al. (2012)	Journal of Technology Transfer
Schoonmaker et al. (2013)	Journal of Technology Transfer
Sedaitis (2000)	Research Policy
Siegel et al. (2003)	Research Policy
Stone (2003)	Journal of Technology Transfer
Swamidass (2013)	Journal of Technology Transfer
Vohora et al. (2004)	Research Policy
Vuola and Hameri (2006)	Technovation
Walter (2000)	Journal of Technology Transfer
Weckowska (2014)	Technovation
Wright et al. (2009)	Journal of Technology Transfer
Zi and Blind (2014)	Journal of Technology Transfer
Ziegler et al. (2013)	Journal of Technology Transfer
Theme: academic entrepreneurship	
Bathelt et al. (2010)	Technovation
Becker and Gassmann (2006)	Journal of Technology Transfer
Beoku-Betts (2005)	Journal of Technology Transfer
Bianchi et al. (2011)	R&D Management
Chapple et al. (2005)	Research Policy
Cooper et al. (2012)	Journal of Technology Transfer
Doganova (2013)	Science and Public Policy
Ferrary (2008)	Journal of Technology Transfer
Festel (2013)	Journal of Technology Transfer
Giaretta (2013)	Journal of Technology Transfer
Goel and Nelson (2009)	Journal of Technology Transfer
Golob (2006)	Journal of Technology Transfer
Guerrero and Urbano (2012)	Journal of Technology Transfer
Guerrero et al. (2014)	Journal of Technology Transfer
Hussler et al. (2010)	Technovation
Jackson and Audretsch (2004)	Journal of Technology Transfer
Johansson et al. (2005)	Journal of Technology Transfer
Kidwell (2014)	Journal of Technology Transfer

Table 2 continued

Author(s)	Journal
Knie and Lengwiler (2008)	Science and Public Policy
Kroll and Liefner (2008)	Technovation
Leitch and Harrison (2005)	R&D Management
Levie (2014)	Journal of Technology Transfer
Libaers (2009)	Journal of Technology Transfer
Link and Siegel (2005)	Research Policy
Maia and Claro (2013)	Journal of Technology Transfer
Markman et al. (2004)	Journal of Technology Transfer
Martinelli et al. (2008)	Journal of Technology Transfer
Moultrie (2015)	Technovation
Nilsson et al. (2010)	Journal of Technology Transfer
Owen-Smith and Powell (2001)	Journal of Technology Transfer
Patton et al. (2009)	Journal of Technology Transfer
Peters et al. (2004)	Journal of Technology Transfer
Saragossi and de la Potterie (2013	Technovation
Rizzo (2014)	Journal of Technology Transfer
Roessner et al. (2010)	Journal of Technology Transfer
Ruan et al. (2014)	Technovation
Salvador (2011)	Journal of Technology Transfer
Sanders and Miller (2010)	Science and Public Policy
Schoen et al. (2012)	Journal of Technology Transfer
Schultz (2011)	Journal of Technology Transfer
Sedaitis (2000)	Research Policy
Siegel et al. (2003)	Research Policy
Swamidass (2013)	Journal of Technology Transfer
Vohora et al. (2004)	Research Policy
Vuola and Hameri (2006)	Technovation
Wright et al. (2009)	Journal of Technology Transfer
Theme: university–industry collaboration	
Acworth (2008)	Research Policy
Boehm and Hogan (2014)	Journal of Technology Transfer
Doganova (2013)	Science and Public Policy
Garrett-Jones et al. (2010)	Journal of Technology Transfer
Kroll and Liefner (2008)	Technovation
Libaers (2009)	Journal of Technology Transfer
Martinelli et al. (2008)	Journal of Technology Transfer
Morandi (2013)	Journal of Technology Transfer
Owen-Smith and Powell (2001)	Journal of Technology Transfer
Saragossi and de la Potterie (2003)	Technovation
Resende et al. (2013)	Technovation
Saragossi and de la Potterie (2003)	Journal of Technology Transfer

Table 2 continued

Author(s)	Journal
Schultz (2011)	Journal of Technology Transfer
Siegel et al. (2003)	Research Policy
Thune and Gulbrandsen (2014)	Journal of Technology Transfer
Theme: Commercialization	
Acworth (2008)	Research Policy
Andersen (2011)	Journal of Technology Transfer
Bianchi et al. (2011)	R&D Management
Boehm and Hogan (2014)	Journal of Technology Transfer
Cho and Lee (2003)	R&D Management
Derrick (2014)	Journal of Technology Transfer
Efstathiades et al. (2000)	Technovation
Festel (2013)	Journal of Technology Transfer
Golob (2006)	Journal of Technology Transfer
Jackson and Audretsch (2004)	Journal of Technology Transfer
Juanola-Feliu et al. (2012)	Technovation
Kidwell (2014)	Journal of Technology Transfer
Kroll and Liefner (2008)	Technovation
Leahy (2003)	Journal of Technology Transfer
Leitch and Harrison (2005)	R&D Management
Levie (2014)	Journal of Technology Transfer
Maia and Claro (2013)	Journal of Technology Transfer
Malik (2002)	Technovation
Matsumoto et al. (2010)	R&D Management
Moncada-Paternò-Castello et al. (2003)	Technovation
Moultrie (2015)	Technovation
Nilsson et al. (2010)	Journal of Technology Transfer
Patton et al. (2009)	Journal of Technology Transfer
Powell and Moris (2004)	Journal of Technology Transfer
Rasmussen et al. (2006)	Technovation
Resende et al. (2013)	Technovation
Ruan et al. (2014)	Technovation
Sanders and Miller (2010)	Science and Public Policy
Saragossi and de la Potterie (2003)	Journal of Technology Transfer
Schoen et al. (2012)	Journal of Technology Transfer
Schoonmaker et al. (2013)	Journal of Technology Transfer
Sedaitis (2000)	Research Policy
Shen (2005)	Science and Public Policy
Stone (2003)	Journal of Technology Transfer
Swamidass (2013)	Journal of Technology Transfer
Weckowska (2014)	Technovation
Ziegler et al. (2013)	Journal of Technology Transfer

Table 2 continued

Author(s)	Journal
Theme: R&D and firm knowledge transfer	
Acworth (2008)	Research Policy
Andersen (2011)	Journal of Technology Transfer
Asheim and Isaksen (2002)	Journal of Technology Transfer
Bach et al. (2002)	Journal of Technology Transfer
Backman et al. (2007)	R&D Management
Bathelt et al. (2010)	Technovation
Becker and Gassmann (2006)	Journal of Technology Transfer
Bessant (1999)	Research Policy
Cho and Lee (2003)	R&D Management
Cruz-Castro et al. (2012)	Journal of Technology Transfer
Daellenbach and Davenport (2004)	Journal of Technology Transfer
Davies and Brady (2000)	Research Policy
Derrick (2014)	Journal of Technology Transfer
Díaz (2012)	Journal of Technology Transfer
Ferrary (2008)	Journal of Technology Transfer
Gallaher and Petrusa (2006)	Journal of Technology Transfer
Garrett-Jones et al. (2010)	Journal of Technology Transfer
Guimén (2009)	Journal of Technology Transfer
Gulbrandsen and Godoe (2008)	Journal of Technology Transfer
Hansen and Ockwell (2014)	Technovation
Hess and Siegwart (2013)	Journal of Technology Transfer
Howells (2006)	Research Policy
Hussler et al. (2010)	Technovation
Intarakumnerd and Charoenporn (2015)	Research Policy
Juanola-Feliu et al. (2012)	Technovation
Kingsley et al. (1996)	Research Policy
Lal (1999)	Research Policy
Leitch and Harrison (2005)	R&D Management
Link and Scott (2004)	Journal of Technology Transfer
Malik (2002)	Technovation
Matsumoto et al. (2010)	R&D Management
Menke et al. (2007)	Journal of Technology Transfer
Moncada-Paternò-Castello et al. (2003)	Technovation
Morandi (2013)	Journal of Technology Transfer
Moultrie (2015)	Technovation
Nightingale (2000)	Research Policy
Ottosson (2004)	Technovation
Peretz and Tonn (2005)	Journal of Technology Transfer
Pittiglio et al. (2009)	Journal of Technology Transfer
Powell and Moris (2004)	Journal of Technology Transfer

Table 2 continued

Author(s)	Journal
Rasmussen et al. (2006)	Technovation
Rizzo (2014)	Journal of Technology Transfer
Roessner et al. (2010)	Journal of Technology Transfer
Ruan et al. (2014)	Technovation
Schoonmaker et al. (2013)	Journal of Technology Transfer
Schultz (2011)	Journal of Technology Transfer
Shen (2005)	Science and Public Policy
Stone (2003)	Journal of Technology Transfer
Vuola and Hameri (2006)	Technovation
Weckowska (2014)	Technovation
Williams-Jones et al. (2014)	Science and Public Policy
Xu et al. (2007)	Journal of Technology Transfer
Youtie et al. (2008)	Journal of Technology Transfer
Zi and Blind (2014)	Journal of Technology Transfer
Ziegler et al. (2013)	Journal of Technology Transfer

Appendix 2

See Table 3.

Table 3 Illustrative working template

#	Article Title	Author 1	Author 1 Affiliations	Author 2	Journal	Year of Publication	Theme	Key Words	Unit of Analysis	Sector	Number of Cases	Main Method	Data Collection	…
1	R&D Venture: proposition of a technology transfer concept for breakthrough technologies with R&D cooperation: A case study in the energy sector	Stephan Hess	Swiss Federal Institute of Technology (ETHZ), Zurich, Switzerland	…	Journal of Technology Transfer	2013	This study investigates if R&D ventures improve the technology transfer for breakthrough innovation to future technology markets	R&D management; Technology transfer; Innovation management; Breakthrough technologies; Technology life cycle management; Corporate venturing	R&D Ventures: Management roles; Power Plants; M&A and R&D management; Academic Partners	Energy	3	Qualitative	Interviews; Workshops; Internal and External Information; R&D reports and documents	…
2	Governance typology of universities' technology transfer processes	Schoen, Anja	TUM School of Mgmt., Technical University of Munich, Germany	…	Journal of technology transfer	2012	This paper presents an analysis of how European universities have set the governance of technology transfer activities and provides a new typology of organizational models	Technology transfer offices; Organizational structure; Governance; Academic patents	Technology Transfer Offices; Universities	Education	16	Qualitative	semi-structured interviews TTO websites; University websites; Office university reports; CEMI Survey	…
3	The trust "builders" in the technology transfer relationships: an Italian science park experience	Giaretta, Elena	Department of Business Administ., University of Verona, Italy	…	Journal of technology transfer	2013	This study proposes and assesses the effectiveness of a solution for disseminating trust in the technology transfer services	Technology transfer; Regional barriers; Trust; Boundary spanner; Small and medium-sized enterprises; Wisdom; Mentoring	Science and technology park; project coordinators; Blue Eagles (retired business owners/ managers)	mechanics-electronics; services; food and agriculture; building; ICT; chemicals-cosmetics-medical equipment; energy, textile; plastic and rubber; furniture manufact.; biotech	30	Qualitative	Questionnaires; Face-to-face interviews; focus groups	…
4	…	…	…	…	…	…	…	…	…	…	…	…	…	…

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